Algorithmics Unit 3

School Assessed Task 2025 – Griddo

Scenario

This task is based on the final question from the 2024 VCE Algorithmics Examination.

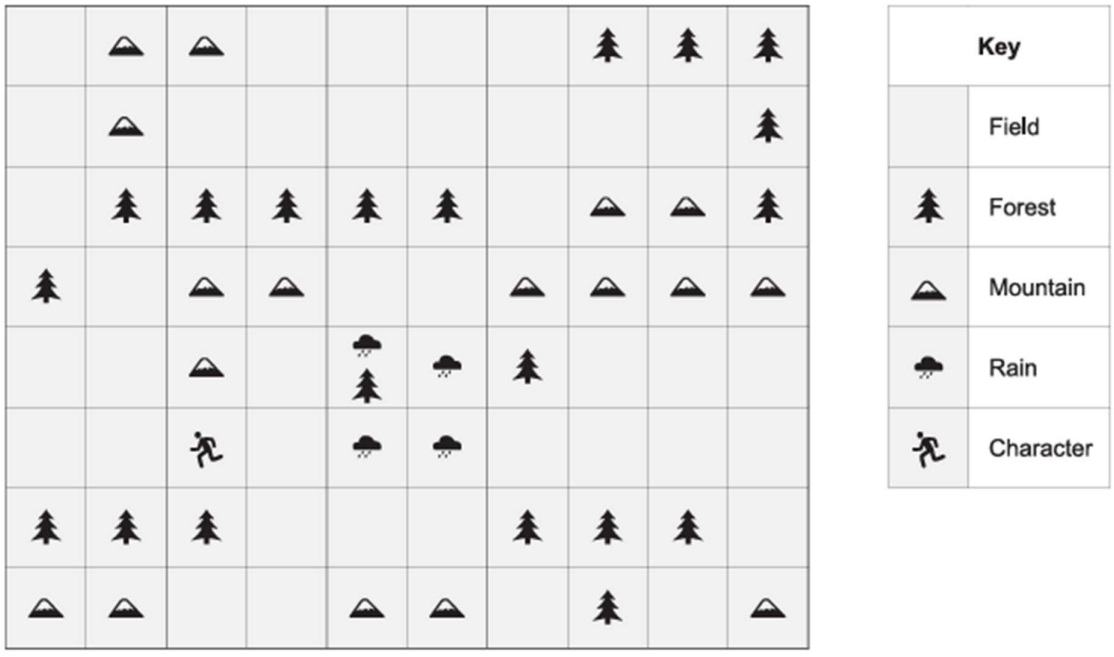
In the computer game 'Griddo', players move characters around a map. The map is a two-dimensional, square grid. Each grid square has a type of terrain.

- Field squares allow for normal movement.
- Forest squares can be moved through, but only at half speed.
- Mountain squares cannot be moved through.

During the game, both field and forest squares may be temporarily modified with rain, halving normal movement speed through that square.

The location of fields, forests and mountains does not change during the game, hence, it is possible to pre-compute the shortest paths between locations on the map without the presence of rain.

An example of a small map is shown below. Maps in the actual game can have millions of squares.



Characters can move up, down, left or right to adjacent squares.

During the game, players can instruct their character to move to a location. When this is done, the game needs to determine the path that the character will take to get there and begin moving the character. It needs to determine the path quickly and also find a fast path for the character to take to get to the destination.